TECHNICAL MEMORANDUM TEST FIXTURE

Verification of a Lunar Thermal Sensor Calibration Method

A. Researcher and B. Engineer

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FICTIONAL TEST FIXTURE—NOT AN OFFICIAL NASA PUBLICATION

Abstract

A repeatable calibration method was evaluated for a fictional lunar thermal sensor over the range from 220 to 320 K. The method reduced the simulated root-mean-square temperature error from 1.8 to 0.4 K. This report records the apparatus, uncertainty assumptions, results, and limitations needed to reproduce the test. All measurements are stated in International System of Units (SI).

Contents

Symbols and Acronyms

T	temperature, K
V	sensor output, V
RMS	root mean square
SI	International System of Units
UTC	coordinated universal time

Chapter 1

Introduction

Thermal instruments used in remote environments require traceable calibration and an explicit treatment of uncertainty. The objective of this fictional study is to verify a second-order voltage-to-temperature conversion over a bounded operating range. The report distinguishes measured values from simulated values and records enough procedural detail for an independent repeat of the analysis.

1.1 Scope

The study covers one engineering unit under steady-state laboratory conditions. Launch loads, radiation damage, and long-duration drift are outside the scope of this fixture.

Chapter 2

Apparatus and Method

The sensor was placed in a simulated thermal-vacuum chamber. Chamber temperature was sampled every 10 s at five plateaus between 220 and 320 K. Voltage was recorded in volts and time was recorded in coordinated universal time (UTC).

The calibration model was

$$T(V) = a_0 + a_1 V + a_2 V^2, \tag{2.1}$$

where the coefficients were estimated by least squares. The uncertainty budget included reference-probe accuracy, voltage quantization, and repeatability.

Chapter 3

Results and Discussion

Table 3.1 summarizes the fictional verification points. Reported precision does not exceed the precision supported by the simulated apparatus.

Table 3.1: Calibration verification results.

Reference temperature, K	Raw error, K	Corrected error, K
220.0	1.7	0.3
270.0	1.9	0.4
320.0	1.8	0.5

The residuals showed no monotonic trend across the tested range. Because the data are simulated, the result demonstrates document and workflow behavior only; it does not establish flight qualification or instrument performance.

Chapter 4

Conclusions

The fictional second-order calibration reduced RMS error over the stated test range. A real qualification campaign would require additional units, cycling, radiation exposure, and an independently reviewed uncertainty analysis.

Appendix A

Calibration Coefficients

The fictional coefficients used to generate the verification data were $a_0 = 198.2$ K, $a_1 = 24.6$ K/V, and $a_2 = 0.82$ K/V^2. These values are test data and must not be used for hardware calibration.

Bibliography

[1] National Aeronautics and Space Administration, *NASA Publications Guide for Authors*, NASA/SP–2005–7602/REV1, 2005.

[2] National Institute of Standards and Technology, *Guide for the Use of the International System of Units (SI)*, NIST Special Publication 811, 2008.

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